An asymptotic preserving scheme for kinetic equations describing bacterial travelling pulses

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Abstract

1 Introduction

Chemotaxis is the directed movement of bacteria in response to chemical fluctuations in the environment. Many bacteria navigate towards favourable conditions by sensing gradients of nutrients or chemoattractants and away from harmful substances. This behaviour is essential for survival [?] and it plays a crucial role in many biological processes such as wound healing, fertilization and inflammation [?], in multicellular organisms, and for unicellular organisms is used as a foraging mechanism [?] [?].

Although bacterial chemotaxis has been studied since the end of the 19th century, it was not until the work of E.F. Keller and L.A. Segel in 1970 that a precise mathematical model was proposed [?]. This result became a cornerstone in the field of mathematical biology and remains one of the most influential and widely cited to this day, thanks to the rigorous coupled partial differential equation (PDE) system introduced for modelling chemotaxis.

The general form of Keller-Segel (K-S) model is

$$\begin{cases} \dfrac{\partial \rho}{\partial t} = \nabla(D_\rho(\rho, S)\nabla\rho - \chi(\rho, S)\nabla S) + f(\rho, S), \\ \dfrac{\partial S}{\partial t} = D_S \Delta S + g(\rho, S) - h(\rho, S)S. \end{cases} \tag{1.1}$$

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Here, $\rho(t,x)$ represents the bacteria density on a given domain $\Omega \subset \mathbb{R}^n$ and $S(t,x)$ denotes the concentration of the chemical signal. The first term in the first equation in (1.1) describes the random component of the movement, while the second term describes the bias component, bacteria move towards high concentrations of the chemical. Consequently, the function $D_\rho(\rho,S)$ describes the diffusivity of the bacteria (random component), and $\chi(\rho,S)$ represents the chemotactic sensitivity, which measures how sensible bacteria are to a chemical variation (biased component). Both D_ρ and χ may be functions of ρ and S. The function $f(\rho,S)$ accounts for bacteria growth and death, whereas the functions $g(\rho,S)$ and $h(\rho,S)$ are functions that describe production and degradation of the chemical signal, respectively. Finally, D_S is a diffusion coefficient of S.

We study the migration of *E. coli* driven by high concentrations of nutrient N and chemoattractant S, motivated by the microchannel experiments in [**?**]. Roughly 10^5 cells are initially concentrated at the left end of a 2 cm channel (width 500 μm, height 100 μm); nutrient is supplied at the right end. On the ~ 100 s time scale, a dense pulse forms and propagates rightward with nearly constant speed and mass. The profile is strongly asymmetric—diffuse at the front and sharp at the rear—and this shape is preserved during propagation. Cell division is neglected because the experimental duration (hours) is short relative to the cell cycle.

We numerically investigate whether the relevant macroscopic system of PDEs corresponds to the underlying physical system described by the kinetic equation in the diffusion limit and to verify the emergence of travelling wave solutions at both the kinetic and macroscopic levels. These methods are widely known as asymptotic preserving schemes (AP) [**?**, **?**] since they mimic the asymptotic behaviour of the kinetic equation when the scaling parameter approaches to zero and the mesh size and time steps are fixed. In this paper, we use a micro-macro decomposition of the unknown in the sense of [**?**] as detailed in Section 4.

Following the seminal works by Alt [**?**], Dunbar and Othmer [**?**] we start from the following kinetic equation

$$\partial_t f(t,x,v) + v\partial_x f(t,x,v) = \int_V \Big(T(t,x,v,v') f(t,x,v') - T(t,x,v',v) f(t,x,v) \Big)\, \mathrm{d}v' \ . \quad (1.2)$$

The bacteria population is described by the mesoscopic density $f(t,x,v)$ where $(x,v) \in \mathbb{R} \times V$ since we are looking for planar travelling wave solutions. Here V denotes a compact interval of admissible velocities. The turning kernel T describes the change in direction during the tumbling phase of the movement. T depends on the space and time variables through the chemical signals in the medium, and $T(\cdot, v, v')$ represents the change of velocity from v' to v. Following [**?**], we consider the influence of two chemical species, $S(t,x)$, a chemotactic signal released by the bacteria, and $N(t,x)$, a nutrient consumed by the bacteria. We assume that both chemical cues influence the tumbling kernel in an additive way as we explain below. For simplicity we are going to assume that T only depends on the current velocity with the condition that $\int_V T(t,x,v)\, \mathrm{d}v = 1$.

Let

$$T_\epsilon(t,x,v) = \frac{\psi_\epsilon(t,x,v)}{\displaystyle\int_V \psi_\epsilon(t,x,v)\,\mathrm{d}v} \ , \tag{1.3}$$

where we assume that the dependence of the turning operator on the chemical concentration S and nutrient N happens as a perturbation of magnitude $\epsilon \ll 1$ in the following way

$$\psi_\epsilon(t,x,v) = 1 + \epsilon\psi_1(t,x,v) \ , \tag{1.4}$$

$$\psi_1(t,x,v) = \phi_1[S](v) + \phi_2[N](v) \ , \quad \int_V \psi_1(t,x,v)\,\mathrm{d}v = 0 \ . \tag{1.5}$$

Here ϕ_1 and ϕ_2 are given by

$$\phi_1[S](v) = \chi_S\phi\Big(\frac{DS}{Dt}\Big|_v\Big) \quad , \quad \phi_2[N](v) = \chi_N\phi\Big(\frac{DN}{Dt}\Big|_v\Big) \ ,$$

and $\frac{D}{Dt} = \partial_t + v\partial_x$ is the total derivative along the direction given by v.

Using (1.4), the turning kernel T_ϵ writes

$$T_\epsilon(t,x,v) = 1 + \epsilon\Big[\chi_S\phi\Big(\frac{DS}{Dt}\Big|_v\Big) + \chi_N\phi\Big(\frac{DN}{Dt}\Big|_v\Big)\Big] + \mathcal{O}(\epsilon^2) \ , \tag{1.6}$$

where the error terms contained in $\mathcal{O}(\epsilon^2)$ integrate to 0 in the velocity variable. We also note that using (1.5) we have $\int_V T_\epsilon(t,x,v)\,\mathrm{d}v = 1$ which describes the conservation of individuals during the velocity reorientation.

Let

$$\widetilde{\psi}_1(t,x,v) = \phi_1[S](v) + \phi_2[N](v),$$

where

$$\phi_1[S](v) = \chi_S\phi\left(\frac{DS}{Dt}\bigg|_v\right), \qquad \phi_2[N](v) = \chi_N\phi\left(\frac{DN}{Dt}\bigg|_v\right),$$

and $\frac{D}{Dt} = \partial_t + v\partial_x$ is the material derivative along the direction v. Since, in general,

$$\int_V \widetilde{\psi}_1(t,x,v)\,dv \neq 0,$$

we introduce its centered version

$$\psi_1(t,x,v) = \widetilde{\psi}_1(t,x,v) - \frac{1}{|V|}\int_V \widetilde{\psi}_1(t,x,w)\,dw,$$

so that

$$\int_V \psi_1(t,x,v)\,dv = 0.$$

We then define the rescaled turning kernel by

$$T_\varepsilon(t,x,v) = |V| \frac{1+\varepsilon\widetilde{\psi}_1(t,x,v)}{\displaystyle\int_V \big(1+\varepsilon\widetilde{\psi}_1(t,x,w)\big)\,dw}.$$

A first-order expansion in ε yields

$$T_\varepsilon(t,x,v) = 1+\varepsilon\psi_1(t,x,v)+O(\varepsilon^2),$$

with

$$\int_V T_\varepsilon(t,x,v)\,dv = |V|.$$

The chemoattractant and nutrients concentrations evolve according to reaction-diffusion equations

$$\partial_t S = D_S \partial_{xx} S - \alpha S + \beta\rho \ , \tag{1.7}$$

$$\partial_t N = D_N \partial_{xx} N - \gamma\rho N \ . \tag{1.8}$$

The chemoattractant S is consumed and produced by the cells with rates given by α and β, respectively. The nutrients are consumed at rate γ. Both, chemoattractant and nutrients diffuse in the medium with constant D_S and D_N. Here ρ denotes the macroscopic density defined as

$$\rho(t,x) = \int_V f(t,x,v)\,\mathrm{d}v \ .$$

As mentioned before, we are going to study this biological phenomenon using an approach distinct from classical modelling. In particular, the biased component in the equation of ρ in (1.1), which is going to be derived from a kinetic description of the motion, will be uniformly bounded, whereas in the case of K-S this biased component is the chemotactic sensitivity function $\chi(\rho,S)$, which is proportional to the gradient of the chemical, is generally unbounded when instability due to aggregation takes place. This will provide us with a more precise and rigorous description of the evolution of baterial movement in this specific context, what is to say, it will allow us to describe analytically the travelling wave behaviour it is observed experimentally.

In this paper we numerically study the macroscopic limit of the kinetic equation (1.2) to the macroscopic PDE

$$\begin{cases} \partial_t \rho = D_\rho \Delta\rho - \nabla\cdot(\rho u_S + \rho u_N), \\ \partial_t S = D_S \Delta S - \alpha S + \beta\rho, \\ \partial_t N = D_N \Delta N - \gamma\rho N. \end{cases} \tag{1.9}$$

On the one hand, the density of bacteria $\rho(t,x)$ (at time $t>0$ and position $x\in\mathbb{R}$) follows a drift-diffusion equation, combining linear diffusion (first of the right hand side) together with transport processes (second term). On the other hand, we consider the chemoattractant signal $S(t,x)$ and the nutrient $N(t,x)$ which both follow reaction-diffusion equations driving the external chemical concentrations. The latter term of the first equation includes u_S and u_N that are fluxes coming from the chemoattractant and the nutrient, respectively, and which are responsible for the biased movement of the bacteria. In particular, the flux u_S contributes to gather the bacterial density and create a pulse, and the flux u_N is responsible for the motion of this pulse towards higher nutrient levels. Finally, we consider that the chemoattractant is naturally degraded at rate α and produced by the bacteria at rate β. Also, the nutrients are consumed at rate γ.

In contrast to the definition of the fluxes in the K-S system, given by

$$u_S = \chi(\rho, S)\nabla S, \qquad u_N = \chi(\rho, N)\nabla N,$$

here we propose to follow the analysis in [?], in which the fluxes are

$$\begin{aligned} u_S &= \chi_S J_\phi(-\epsilon\partial_t S, |\nabla S|)\frac{\nabla S}{|\nabla S|}, \\ u_N &= \chi_N J_\phi(-\epsilon\partial_t N, |\nabla N|)\frac{\nabla N}{|\nabla N|}, \end{aligned} \tag{1.10}$$

based on the temporal response of bacteria ϕ. We refer to Section 2 for a detailed definition of this function. This ϕ is called a response function, and describes how a single bacteria responds to a change in the concentration of the chemoattractant S (or of the nutrient N) in the surrounding environment. The function J_ϕ incorporates the microscopic features that derive from the concrete response of each bacterium ϕ to a change in the environment. The parameter ϵ is the ratio between the pulse speed and the speed of individual bacteria, in the experiments we can see that $\epsilon \approx 0.1$, which means that the pulse travels one order of magnitude slower than each individual. Moreover, the constants χ_S and χ_N are the chemotactic biases, which characterise the amplitude of the signal integration function ϕ. Finally, it is worth noting that the fluxes u_S and u_N are uniformly bounded, as we were expecting.

This paper is organised as follows. (...)

2 Macroscopic system of equations

In this section we are going to derive the macroscopic system that is obtained from (1.2) following [?]. We introduce two relevant parameters: the maximum speed of a single bacterium $c = \max\{|v| : v \in V\}$ and the mean turning frequency λ_0.

The main unknown is the speed of the travelling pulse, σ. We start by rescaling the kinetic model (1.2) into a non-dimensional form using the non-dimensional variables, denoted with tildes, as

$$\tilde{t} = \frac{t}{\bar{t}}, \quad \tilde{x} = \frac{x}{\bar{x}}, \quad \tilde{V} = \frac{V}{c}, \quad \tilde{T} = \frac{T}{\lambda_0},$$

to model the travelling wave with $\bar{x} = \sigma\bar{t}$. According to the experiments, the pulse speed σ is much lower than the speed of a single bacterium c, and this difference is measured by the ratio $\epsilon = \sigma/c$, where $\epsilon \ll 1$.

The kinetic equation can be written in the following form, where we drop the tilde for simplicity of notation

$$\epsilon\partial_t f^\epsilon + v\partial_x f^\epsilon = \frac{\mu}{\epsilon}\int_V T_\epsilon(t,x,v')f^\epsilon(t,x,v')\,\mathrm{d}v' - |V|T_\epsilon(t,x,v)f^\epsilon(t,x,v)\ , \tag{2.1}$$

where $\mu/\epsilon = \lambda_0\bar{x}/c$ and with velocity space $V = [-1,1]$ (one-dimensional case). The turning kernel is $T_\epsilon(t,x,v) = 1 + \epsilon\psi_1(t,x,v)$The turning kernel is written in the form

$$T_\varepsilon(t,x,v) = 1 + \varepsilon\psi_1(t,x,v) + O(\varepsilon^2),$$

where ψ_1 denotes the centered first-order perturbation defined above. We now consider a second order regular expansion of f^ϵ in ϵ,

$$f^\epsilon(t,x,v) = f^0(t,x,v) + \epsilon f^1(t,x,v) + \epsilon^2 f^2(t,x,v) + \mathcal{O}(\epsilon^3)\ ,$$

where $\int_V f^\epsilon(t,x,v)\,\mathrm{d}v = \int_V f^0(t,x,v)\,\mathrm{d}v = \rho(t,x)$, therefore $\int_V f^i(t,x,v)\,\mathrm{d}v = 0$, $\forall i \geq 1$.

Introducing this ansatz in (2.1), we obtain

$$\epsilon^0:\quad f^0(t,x,v) = \frac{\rho(t,x)}{|V|}\ , \tag{2.2}$$

$$\epsilon^1:\quad v\partial_x f^0 = \mu\left(\int_V \psi_1(v')f^0(v')\,dv' - |V|\psi_1(v)f^0(v) - |V|f^1(v)\right)\ , \tag{2.3}$$

$$\epsilon^2:\quad \partial_t f^0 + v\partial_x f^1 = \mu\Big(\int_V \psi_1(v')f^1(v')\,\mathrm{d}v' - |V|\psi_1(v)f^1(v) - |V|f^2(v)\Big)\ . \tag{2.4}$$

Next, after replacing f^0 by its expression (Eq. (2.2)), we multiply (2.3) by v and we integrate again with respect to v to obtain

$$\begin{aligned}\mu|V|\int_V vf^1(t,x,v)\,\mathrm{d}v = &-\partial_x\frac{\rho(t,x)}{|V|}\int_V |v|^2\,\mathrm{d}v - \mu\rho(t,x)\chi_S\int_V \phi(\epsilon\partial_t S + v\partial_x S)v\,\mathrm{d}v\\ &- \mu\rho(t,x)\chi_N\int_V \phi(\epsilon\partial_t N + v\partial_x N)v\,\mathrm{d}v.\end{aligned} \tag{2.5}$$

We define

$$D_\rho = \frac{1}{\mu|V|^2}\int_V v^2\ \mathrm{d}v, \qquad u_S = -\frac{\chi_S}{|V|}\int_V \phi(\varepsilon\partial_t S + v\partial_x S)\,v\ \mathrm{d}v, \qquad u_N = -\frac{\chi_N}{|V|}\int_V \phi(\varepsilon\partial_t N + v\partial_x N)\,v\ \mathrm{d}v. \tag{2.6}$$

In the limiting case where the internal response function is

$$\phi(Y) = -\phi_0\,\mathrm{sign}(Y),$$

the flux terms reduce, up to the constant factor determined by ϕ_0 and the velocity measure, to

$$u_S = \chi_S \, \mathrm{sign}(\partial_x S), \qquad u_N = \chi_N \, \mathrm{sign}(\partial_x N).$$

Finally, integrating (2.4) in v and using (2.6) we obtain

$$\partial_t \rho = D_\rho \partial_{xx} \rho - \partial_x(\rho u_S + \rho u_N) \ .$$

The diffusion coefficient is determined by the velocity distribution and the turning frequency as in (2.6). In particular, when the normalized scaling $\mu|V| = 1$ is used, this coefficient becomes

$$D_\rho = \frac{1}{|V|} \int_V v^2 \, dv. \tag{2.7}$$

For example, when $V = [-1, 1]$, we have $D_\rho = \frac{1}{3}$. We refer to [?, ?] for more details of the derivation.

3 Existence of travelling wave solutions (not sure if this is necessary)

In this section we summarise the results obtained in [?] for the existence of travelling wave solutions for the kinetic equation (1.2), and for the macroscopic system (1.9).

Consider a travelling wave solution $z = x - ct$ where c denotes the speed of the wave. We rewrite (1.2) in the moving frame variable z, (σ is used for wave speed in other sections)

$$\begin{aligned} (v - c)\partial_z f(z, v) &= \int T(z, v' - c) f(z, v') d\nu(v') - T(z, v - c) f(z, v) \ , \\ -c\partial_z S(z) &= D_S \partial_z^2 S(z) - \alpha S(z) + \beta \rho(z) \ , \\ -c\partial_z N(z) &= D_N \partial_z^2 N(z) - \gamma \rho(z) N(z) \ , \end{aligned} \tag{3.1}$$

where ν is a symmetric probability measure on $\mathbb{R}^d$, where $V = \mathrm{supp}\ \nu$. Throughout this section we assume that ν is absolutely continuous with respect to the Lebesgue measure. The probability density function belongs to L^p for some $p > 1$, i.e.

$$\mathrm{d}\nu(v) = w(v) \, \mathrm{d}v \ , \quad w \in L^p \ , \quad p \in (1, \infty] \ .$$

We assume that the the fresh nutrient is located on the right so the wave moves towards high concentrations of the nutrient, i.e. $\partial_z N > 0$. We expect the chemoattractant concentration to have a maximum at the cell density peak, and therefore we look for a solution where $\partial_z S$ changes sign. Without loss of generality we assume that this maximal point is at $z = 0$. We summarise these conditions as follows

$$\begin{aligned} & S(\pm\infty) = 0 \ , \quad N(\pm\infty) = N_\pm \ , \\ & \partial_z S > 0 \quad \text{for} \quad z < 0 \ , \quad \text{and} \quad \partial_z S < 0 \quad \text{for} \quad z > 0 \ , \\ & \partial_z N > 0 \ , \quad \forall z \in \mathbb{R} \ . \end{aligned} \tag{3.2}$$

Moreover, the tumbling rate T in (3.1) can take only four values depending on the sign of the gradients,

$$T(z, v-c) = 1 - \chi_S \mathrm{sign}((v-c)\partial_z S(z)) - \chi_N \mathrm{sign}((v-c)\partial_z N(z)) \;, \tag{3.3}$$

Macroscopic travelling pulse We present a mathematical description of travelling pulses at the macroscopic scale. For the proof of this theorem we refer the reader to [**?**].

Theorem 3.1 **([?])** *There exist a speed $c > 0$, and two positive values $N_- < N_+$, such that the system (1.9) admits travelling wave solutions $(\rho(x-ct), S(x-ct), N(x-ct))$ with the following boundary conditions:*

$$\lim_{|z|\to\infty} \rho = \lim_{|z|\to\infty} S = 0, \qquad \lim_{z\to-\infty} N = N_- < N_+ = \lim_{z\to+\infty} N.$$

Assume in addition that $\partial_z S$ changes sign only once, and that $\partial_z N$ is positive everywhere. Then, the speed $c > 0$ is uniquely determined by the following implicit relation,

$$\chi_N - \frac{c}{1-(\epsilon c)^2} = \chi_S \frac{c}{\sqrt{4D_S\alpha + c^2}}.$$

The wave speed formula does not match the travelling speed from the numerical solution of the kinetic model.

For the proof of this theorem we assume that the chemoattractant S has a unique maximum at $z = 0$ and that the nutrient N is increasing. Then, this initial ansatz allows us to decouple the system of equations (1.9), enabling us to solve for ρ, and consequently for S. Finally, we will validate the ansatz, which will lead to the equation for c as stated in Theorem 3.1.

Kinetic travelling pulse It is natural to expect that some of the qualitative behaviours observed at the macroscopic scale, such as the existence of travelling pulse solutions, should also be present at the mesoscopic level. This motivates the study of the existence of travelling wave solutions directly within the mesoscopic framework, aiming to confirm that such wave-like structures are inherent to the dynamics of bacterial motion, regardless of the scale under consideration. In the following, we summarise the results in [**?**] for completeness.

The space variable of the moving pulse at speed c is denoted by $z = x - ct$. We use the following convention: the superscripts $^+$ and $^-$ denote the position of the velocity v with respect to the speed c: for $v > c$ and for $v < c$, respectively. Whereas the subscripts $_+$ and $_-$ denote the relative position with respect to the origin: for $z > 0$ and for $z < 0$, respectively.

Similarly, we denote the spatial density of the same group of bacteria $f_+^-(z, v)$ as

$$\rho_+^-(z) = \int_{v<c} f_+(z, v) d\nu(v).$$

Moreover, the tumbling rate can take only four values $1 \pm \chi_S \pm \chi_N$ denoted by $T_+^+, T_+^-, T_-^+, T_-^-$ which depend on weather v crosses c or weather z crosses the origin or not. We illustrate this convention in Figure 6 in Appendix A.

Therefore, $T \in [1-\chi_S-\chi_N, 1+\chi_S+\chi_N]$ and we conclude that the tumbling frequency is higher when bacteria move in the favourable direction and smaller when bacteria move in the unfavourable direction.

Theorem 3.2 ([?]) *Assume $(\chi_S, \chi_N) \in (0, 1/2) \times [0, 1/2)$. Under the assumption (A1), and conditions*

$$\frac{c_* + \sqrt{(c_*)^2 + 4\alpha D_S}}{c_* + \sqrt{(c_*)^2 + 4\alpha D_S} + 2D_S\lambda_-(c_*)} \leq \textit{explicit constant}, \tag{3.4}$$

$$\frac{-c^* + \sqrt{(c^*)^2 + 4\alpha D_S}}{-c^* + \sqrt{(c^*)^2 + 4\alpha D_S} + 2D_S\lambda_+(c^*)} \leq \textit{explicit constant}, \tag{3.5}$$

$$\textit{Either } \chi_N \geq \chi_S \quad \textit{or} \quad \frac{\sqrt{\alpha/D_S} + \lambda_+(0)}{\sqrt{\alpha/D_S} + \lambda_-(0)} \leq \textit{explicit constant}, \tag{3.6}$$

where $\lambda_-(0)$ and $\lambda_+(0)$ are the smallest positive roots of, respectively

$$\int \frac{v}{T_-(v) + \lambda v} d\nu(v) = 0, \qquad \int \frac{v}{T_+(v) - \lambda v} d\nu(v) = 0.$$

There exist a non negative c, and positive functions $(f, S, N) \in (L^1 \cap L^\infty(\mathbb{R} \times V)) \times \mathcal{C}^2(\mathbb{R}) \times \mathcal{C}^2(\mathbb{R})$ solution of the travelling wave problem,

$$\begin{cases} (v-c)\partial_z f(z,v) = \displaystyle\int T(z, v'-c) f(z,v') d\nu(v') - T(z, v-c) f(z,v), \\ \hfill -c\partial_z S(z) - D_S \partial_z^2 S(z) + \alpha S(z) = \beta\rho(z), \\ \hfill -c\partial_z N(z) - D_N \partial_z^2 N(z) = -\gamma\rho(z) N(z), \end{cases} \tag{3.7}$$

where the tumbling rate T is given by

$$T(z, v-c) = 1 - \chi_S \operatorname{sign}((v-c)\partial_z S(z)) - \chi_N \operatorname{sign}((v-c)\partial_z N(z)).$$

The details of the proof of this theorem can be found in [?].
An important issue discussed in [?] is regarding the uniqueness of the travelling wave, where the authors numerically observed that travelling wave exists for several values of c, in a parameter regime which violates conditions (3.4)-(3.6). This clearly contradicts the results observed at the macroscopic level, where there exists a unique wave speed.

(this is something we should look into in our AP scheme)

4 Micro-macro decomposition

Throughout this section, we use the normalized micro-macro formulation obtained under the scaling $\mu|V| = 1$. To construct an asymptotic preserving scheme consistent with the diffusive limit, we introduce the equilibrium distribution

$$M(v) = \frac{1}{|V|}, \qquad \int_V M(v)\,dv = 1.$$

Since the leading-order term is given by $f^0 = \rho/|V|$, we decompose the kinetic density as

$$f^\varepsilon(t,x,v) = \frac{\rho^\varepsilon(t,x)}{|V|} + \varepsilon g^\varepsilon(t,x,v), \qquad \int_V g^\varepsilon(t,x,v)\,dv = 0. \tag{4.1}$$

We denote by $\langle \cdot \rangle$ the integration with respect to v over V, namely

$$\langle h \rangle = \int_V h(v)\,dv,$$

and define the projection operator

$$\Pi h = \langle h \rangle M.$$

Then $(I - \Pi)g^\varepsilon = g^\varepsilon$.

Integrating (2.1) with respect to v, and using the conservation property of the collision operator together with

$$\langle vM \rangle = \frac{1}{|V|}\int_V v\,dv = 0,$$

we obtain the macroscopic equation

$$\partial_t \rho^\varepsilon + \partial_x \langle v g^\varepsilon \rangle = 0.$$

Next, we define the collision operator

$$\mathcal{Q}_\varepsilon(f) = \int_V T_\varepsilon(t,x,v') f(v')\,dv' - |V| T_\varepsilon(t,x,v) f(v).$$

The normalized turning kernel admits the first-order expansion

$$T_\varepsilon(t,x,v) = 1 + \varepsilon\psi_1(t,x,v) + O(\varepsilon^2), \qquad \langle \psi_1 \rangle = 0.$$

In the following construction, we use the first-order approximation

$$T_\varepsilon^{\mathrm{app}}(t,x,v) = 1 + \varepsilon\psi_1(t,x,v).$$

It preserves the normalization

$$\langle T_\varepsilon^{\mathrm{app}} \rangle = |V|.$$

For the approximate collision operator

$$\mathcal{Q}_\varepsilon^{\text{app}}(f) = \int_V T_\varepsilon^{\text{app}}(t,x,v')f(v')\,dv' - |V|T_\varepsilon^{\text{app}}(t,x,v)f(v),$$

using (4.1), we obtain

$$(I-\Pi)\mathcal{Q}_\varepsilon^{\text{app}}(f^\varepsilon) = -\varepsilon\big(|V|g^\varepsilon + \rho^\varepsilon\psi_1\big) - \varepsilon^2|V|(I-\Pi)(\psi_1 g^\varepsilon).$$

Applying $I-\Pi$ to the kinetic equation with $\mathcal{Q}_\varepsilon^{\text{app}}$ gives

$$\varepsilon^2\partial_t g^\varepsilon + \varepsilon(I-\Pi)(v\partial_x g^\varepsilon) + vM\,\partial_x\rho^\varepsilon = -\mu\big(|V|g^\varepsilon + \rho^\varepsilon\psi_1\big) - \mu\varepsilon|V|(I-\Pi)(\psi_1 g^\varepsilon).$$

Using $M = 1/|V|$ together with the normalization $\mu|V| = 1$, we obtain the micro-macro system

$$\begin{cases}\partial_t\rho^\varepsilon + \partial_x\langle v g^\varepsilon\rangle = 0,\\[4pt] \partial_t g^\varepsilon + \dfrac{1}{\varepsilon}(I-\Pi)(v\partial_x g^\varepsilon) = \dfrac{1}{\varepsilon^2}\left(-\dfrac{v}{|V|}\partial_x\rho^\varepsilon - g^\varepsilon - \dfrac{\rho^\varepsilon}{|V|}\psi_1\right) - \dfrac{1}{\varepsilon}(I-\Pi)(\psi_1 g^\varepsilon).\end{cases} \tag{4.2}$$

5 An asymptotic preserving finite difference scheme

In this section we construct an asymptotic preserving finite difference scheme for the micro-macro formulation (4.2), which is obtained from the first-order approximation of the turning kernel. We present the method in one space dimension. The scheme is designed to remain stable as $\varepsilon \to 0$ and to reduce to a consistent discretization of the macroscopic drift-diffusion limit.

Let $x_{j+\frac12} = x_{\min} + (j+\frac12)\Delta x$, $x_j = x_{\min} + j\Delta x$, and $t_n = n\Delta t$. We approximate ρ, S, and N at the half grid points, while the microscopic unknown g is approximated at the integer grid points,

$$\rho^n_{j+\frac12} \approx \rho(t_n, x_{j+\frac12}), \qquad S^n_{j+\frac12} \approx S(t_n, x_{j+\frac12}), \qquad N^n_{j+\frac12} \approx N(t_n, x_{j+\frac12}),$$

and

$$g^n_{j,k} \approx g(t_n, x_j, v_k), \qquad v_k \in V_h.$$

For a half-grid scalar variable $q^n_{j+\frac12}$, we use

$$\delta_t^+ q^n_{j+\frac12} = \frac{q^{n+1}_{j+\frac12} - q^n_{j+\frac12}}{\Delta t}, \qquad (\delta_x q)^n_j = \frac{q^n_{j+\frac12} - q^n_{j-\frac12}}{\Delta x},$$

and

$$(\delta_x^2 q)^n_{j+\frac12} = \frac{q^n_{j+\frac32} - 2q^n_{j+\frac12} + q^n_{j-\frac12}}{\Delta x^2}.$$

For the integer-grid functions F_j and $g_{j,k}^n$, we define the discrete divergence by

$$(\delta_x F)_{j+\frac{1}{2}} = \frac{F_{j+1} - F_j}{\Delta x}, \quad (\delta_x g)_{j+\frac{1}{2},k}^n = \frac{g_{j+1,k}^n - g_{j,k}^n}{\Delta x}.$$

Whenever a half-grid variable is needed at an integer grid point, we use the local average

$$q_j^n = \frac{1}{2}\left(q_{j-\frac{1}{2}}^n + q_{j+\frac{1}{2}}^n\right).$$

The value $(\delta_x^2 q)_j^n$ is defined in the same way by averaging the two neighboring half-grid values.

Let $V_h = \{v_k\}$ be a symmetric discrete velocity set. We define

$$\langle \eta_k \rangle_h = \Delta v \sum_k \eta_k, \qquad |V|_h = \langle 1 \rangle_h, \qquad \Pi_h \eta_k = \frac{\langle \eta_k \rangle_h}{|V|_h}.$$

We assume

$$\langle v_k \rangle_h = 0.$$

Thus $I - \Pi_h$ is the projection onto the discrete zero-average space in the velocity variable.

Discrete response functions At each time level, the response functions are evaluated explicitly. Since

$$D_t^v S = \partial_t S + v\partial_x S, \qquad D_t^v N = \partial_t N + v\partial_x N,$$

we approximate the time derivatives by using the equations for S and N at time level n. We define

$$\mathcal{L}_{S,j}^n = \widetilde{D}_S(\delta_x^2 S)_j^n - \widetilde{\alpha} S_j^n + \widetilde{\beta}\rho_j^n, \quad \mathcal{L}_{N,j}^n = \widetilde{D}_N(\delta_x^2 N)_j^n - \widetilde{\gamma}\rho_j^n N_j^n.$$

The discrete material derivatives are then given by

$$D_{t,h}^{v_k} S_j^n = \mathcal{L}_{S,j}^n + v_k(\delta_x S)_j^n, \quad D_{t,h}^{v_k} N_j^n = \mathcal{L}_{N,j}^n + v_k(\delta_x N)_j^n.$$

All quantities on the right-hand sides are known at time level n.

The velocity average of $\phi(a + bv)$ does not vanish in general, even when ϕ is odd. We therefore use the normalized discrete responses

$$\psi_{S,j,k}^n = \chi_S \left[\phi\left(D_{t,h}^{v_k} S_j^n\right) - \Pi_h \phi\left(D_{t,h}^{v_k} S_j^n\right)\right],$$

and

$$\psi_{N,j,k}^n = \chi_N \left[\phi\left(D_{t,h}^{v_k} N_j^n\right) - \Pi_h \phi\left(D_{t,h}^{v_k} N_j^n\right)\right].$$

We set

$$\psi_{j,k}^n = \psi_{S,j,k}^n + \psi_{N,j,k}^n.$$

By construction,

$$\langle \psi^n_{S,j,k} \rangle_h = 0, \qquad \langle \psi^n_{N,j,k} \rangle_h = 0, \qquad \langle \psi^n_{j,k} \rangle_h = 0.$$

Consequently, the discrete turning kernel

$$T^n_{\varepsilon,j,k} = 1 + \varepsilon \psi^n_{j,k}$$

satisfies

$$\langle T^n_{\varepsilon,j,k} \rangle_h = |V|_h.$$

We also introduce

$$B^n_{S,j} = \langle v_k \psi^n_{S,j,k} \rangle_h, \qquad B^n_{N,j} = \langle v_k \psi^n_{N,j,k} \rangle_h.$$

The corresponding discrete drift velocities are defined by

$$u^n_{S,j} = -\frac{B^n_{S,j}}{|V|_h}, \qquad u^n_{N,j} = -\frac{B^n_{N,j}}{|V|_h}.$$

Fully discrete micro-macro scheme The fully discrete AP scheme is given by

$$\begin{cases} \delta_t^+ \rho^n_{j+\frac{1}{2}} + \left\langle v_k (\delta_x g)^{n+1}_{j+\frac{1}{2},k} \right\rangle_h = 0, \\ \delta_t^+ g^n_{j,k} + \dfrac{1}{\varepsilon}(I - \Pi_h)\mathcal{A}^n_{j,k} = \dfrac{1}{\varepsilon^2}\left(K^{n,n+1}_{j,k} - g^{n+1}_{j,k} \right) - \dfrac{1}{\varepsilon}(I - \Pi_h)(\psi^n_{j,k} g^n_{j,k}), \\ \delta_t^+ S^n_{j+\frac{1}{2}} = \widetilde{D}_S (\delta_x^2 S)^n_{j+\frac{1}{2}} - \widetilde{\alpha} S^n_{j+\frac{1}{2}} + \widetilde{\beta} \rho^n_{j+\frac{1}{2}}, \\ \delta_t^+ N^n_{j+\frac{1}{2}} = \widetilde{D}_N (\delta_x^2 N)^n_{j+\frac{1}{2}} - \widetilde{\gamma} \rho^n_{j+\frac{1}{2}} N^n_{j+\frac{1}{2}}. \end{cases} \tag{5.1}$$

Here the microscopic transport is treated explicitly by the upwind approximation

$$\mathcal{A}^n_{j,k} = v_k^+ (\delta_x g)^n_{j-\frac{1}{2},k} - v_k^- (\delta_x g)^n_{j+\frac{1}{2},k},$$

where

$$v_k^+ = \max\{v_k, 0\}, \qquad v_k^- = \max\{-v_k, 0\}.$$

The local equilibrium part is defined by

$$K^{n,n+1}_{j,k} = -\frac{v_k}{|V|_h} (\delta_x \rho)^{n+1}_j - \frac{\rho^{[S],n+1}_j}{|V|_h} \psi^n_{S,j,k} - \frac{\rho^{[N],n+1}_j}{|V|_h} \psi^n_{N,j,k}. \tag{5.2}$$

The two upwind values are chosen according to the effective velocities $u^n_{S,j}$ and $u^n_{N,j}$,

$$\rho^{[S],n+1}_j = \begin{cases} \rho^{n+1}_{j-\frac{1}{2}}, & u^n_{S,j} \ge 0, \\ \rho^{n+1}_{j+\frac{1}{2}}, & u^n_{S,j} < 0, \end{cases} \quad \rho^{[N],n+1}_j = \begin{cases} \rho^{n+1}_{j-\frac{1}{2}}, & u^n_{N,j} \ge 0, \\ \rho^{n+1}_{j+\frac{1}{2}}, & u^n_{N,j} < 0. \end{cases}$$

This choice is consistent with the drift part of the limiting flux since

$$-\frac{B^n_{S,j}}{|V|_h}\rho_j^{[S],n+1} - \frac{B^n_{N,j}}{|V|_h}\rho_j^{[N],n+1} = u^n_{S,j}\rho_j^{[S],n+1} + u^n_{N,j}\rho_j^{[N],n+1}.$$

The scheme preserves the microscopic constraint. Indeed, taking the discrete velocity average of the second equation in (5.1) and using

$$\langle K^{n,n+1}_{j,k}\rangle_h = 0$$

gives

$$\langle g^{n+1}_{j,k}\rangle_h = \frac{\varepsilon^2}{\varepsilon^2+\Delta t}\langle g^n_{j,k}\rangle_h.$$

Thus $\langle g^n_{j,k}\rangle_h = 0$ is propagated in time if it holds initially.

Decoupled solution strategy Following the micro-macro strategy in [**?**], the scheme can be solved in a decoupled way. We first introduce an auxiliary unknown $\widetilde{g}^{n+1}_{j,k}$ by

$$\frac{\widetilde{g}^{n+1}_{j,k} - g^n_{j,k}}{\Delta t} + \frac{1}{\varepsilon}(I-\Pi_h)\mathcal{A}^n_{j,k} = \frac{1}{\varepsilon^2}\left(\widetilde{K}^n_{j,k} - \widetilde{g}^{n+1}_{j,k}\right) - \frac{1}{\varepsilon}(I-\Pi_h)(\psi^n_{j,k}g^n_{j,k}). \tag{5.3}$$

where

$$\widetilde{K}^n_{j,k} = -\frac{v_k}{|V|_h}(\delta_x\rho)^n_j - \frac{\rho_j^{[S],n}}{|V|_h}\psi^n_{S,j,k} - \frac{\rho_j^{[N],n}}{|V|_h}\psi^n_{N,j,k}.$$

The upwind values $\rho_j^{[S],n}$ and $\rho_j^{[N],n}$ are defined in the same way as above, with ρ^{n+1} replaced by ρ^n.

All terms in (5.3) are known except $\widetilde{g}^{n+1}_{j,k}$. Hence

$$\widetilde{g}^{n+1}_{j,k} = \frac{\varepsilon^2}{\varepsilon^2+\Delta t}g^n_{j,k} - \frac{\varepsilon\Delta t}{\varepsilon^2+\Delta t}(I-\Pi_h)\mathcal{A}^n_{j,k} + \frac{\Delta t}{\varepsilon^2+\Delta t}\widetilde{K}^n_{j,k} - \frac{\varepsilon\Delta t}{\varepsilon^2+\Delta t}(I-\Pi_h)(\psi^n_{j,k}g^n_{j,k}).$$

Comparing (5.3) with the second equation in (5.1), we obtain

$$g^{n+1}_{j,k} = \widetilde{g}^{n+1}_{j,k} + \lambda_\varepsilon\left(K^{n,n+1}_{j,k} - \widetilde{K}^n_{j,k}\right), \tag{5.4}$$

where

$$\lambda_\varepsilon = \frac{\Delta t}{\varepsilon^2+\Delta t}.$$

Substituting (5.4) into the first equation of (5.1), we obtain a closed equation for ρ^{n+1}. Define

$$D_{\rho,h} = \frac{\langle v_k^2\rangle_h}{|V|_h}, \qquad b^n_{S,j} = \frac{B^n_{S,j}}{|V|_h}, \qquad b^n_{N,j} = \frac{B^n_{N,j}}{|V|_h}.$$

We also set

$$a^n = \lambda_\varepsilon D_{\rho,h}, \qquad c^n_{S,j} = \lambda_\varepsilon b^n_{S,j}, \qquad c^n_{N,j} = \lambda_\varepsilon b^n_{N,j}.$$

Then the density equation can be written as

$$\frac{\rho^{n+1}_{j+\frac{1}{2}}}{\Delta t} - \delta_x \left(a^n \delta_x \rho^{n+1}\right)_{j+\frac{1}{2}} - \delta_x \left(c^n_S \rho^{[S],n+1}\right)_{j+\frac{1}{2}} - \delta_x \left(c^n_N \rho^{[N],n+1}\right)_{j+\frac{1}{2}} = r^n_{j+\frac{1}{2}}, \tag{5.5}$$

where the known right-hand side is

$$r^n_{j+\frac{1}{2}} = \frac{\rho^n_{j+\frac{1}{2}}}{\Delta t} - \left\langle v_k (\delta_x \widetilde{g})^{n+1}_{j+\frac{1}{2},k} \right\rangle_h - \delta_x \left(a^n \delta_x \rho^n\right)_{j+\frac{1}{2}} - \delta_x \left(c^n_S \rho^{[S],n}\right)_{j+\frac{1}{2}} - \delta_x \left(c^n_N \rho^{[N],n}\right)_{j+\frac{1}{2}}.$$

Thus only a scalar linear system for ρ^{n+1} needs to be solved at each time step. With the above upwind choice for the drift terms, the resulting matrix is tridiagonal. Once ρ^{n+1} is obtained, g^{n+1} is recovered explicitly from (5.4). Finally, S^{n+1} and N^{n+1} are updated by the last two equations in (5.1).

5.1 Asymptotic preserving property

We now verify the formal asymptotic preserving property of the scheme. Let Δt and Δx be fixed. Since

$$\lambda_\varepsilon = \frac{\Delta t}{\varepsilon^2 + \Delta t},$$

λ_ε converges to 1 as ε goes to zero. Moreover, by the definition of $\widetilde{g}^{n+1}$, $\widetilde{g}^{n+1}_{j,k}$ converges to $\widetilde{K}^n_{j,k}$. Using the expression of $\widetilde{K}^n$, we obtain $\left\langle v_k (\delta_x \widetilde{g})^{n+1}_{j+\frac{1}{2},k} \right\rangle_h$ converges to

$$-\delta_x \left(D_{\rho,h} \delta_x \rho^n\right)_{j+\frac{1}{2}} - \delta_x \left(b^n_S \rho^{[S],n}\right)_{j+\frac{1}{2}} - \delta_x \left(b^n_N \rho^{[N],n}\right)_{j+\frac{1}{2}}.$$

It follows from the definition of $r^n_{j+\frac{1}{2}}$ in (5.5) that $r^n_{j+\frac{1}{2}}$ converges to $\frac{\rho^n_{j+\frac{1}{2}}}{\Delta t}$. Therefore, in the limit as ε goes to zero, (5.5) reduces to

$$\frac{\rho^{n+1}_{j+\frac{1}{2}} - \rho^n_{j+\frac{1}{2}}}{\Delta t} - \delta_x \left(D_{\rho,h} \delta_x \rho^{n+1}\right)_{j+\frac{1}{2}} - \delta_x \left(b^n_S \rho^{[S],n+1}\right)_{j+\frac{1}{2}} - \delta_x \left(b^n_N \rho^{[N],n+1}\right)_{j+\frac{1}{2}} = 0. \tag{5.6}$$

This is a consistent implicit finite difference discretization of the limiting drift-diffusion equation

$$\partial_t \rho - \partial_x \left(D_{\rho,h} \partial_x \rho\right) - \partial_x \left(b_S \rho\right) - \partial_x \left(b_N \rho\right) = 0.$$

Since $u_S = -b_S$ and $u_N = -b_N$, the drift contribution may equivalently be written as

$$\partial_x (u_S \rho) + \partial_x (u_N \rho).$$

Thus, for fixed Δt and Δx, the fully discrete scheme reduces to a consistent discretization of the macroscopic drift-diffusion limit as ε goes to zero. This proves the formal asymptotic preserving property in the diffusive regime.

5.2 Bound-preserving property

We now discuss the bound-preserving property of the limiting scheme. In general, the full kinetic scheme is not automatically positivity preserving. In the computation, this issue is handled by a post-processing limiter. For the limiting macroscopic scheme, however, a positivity and boundedness result can be established.

In this subsection, we consider the limiting macroscopic scheme with homogeneous no-flux boundary conditions, so that the discrete boundary fluxes vanish.

Recall that the limiting scheme reads

$$\delta_t^+ \rho_{j+\frac{1}{2}}^n = D_{\rho,h}\delta_x^2 \rho_{j+\frac{1}{2}}^{n+1} - \delta_x \left(u_S^n \rho^{[S],n+1}\right)_{j+\frac{1}{2}} - \delta_x \left(u_N^n \rho^{[N],n+1}\right)_{j+\frac{1}{2}},$$

and the upwind values are chosen according to $u_{S,j}^n$ and $u_{N,j}^n$:

$$\rho_j^{[S],n+1} = \begin{cases} \rho_{j-\frac{1}{2}}^{n+1}, & u_{S,j}^n \geq 0, \\ \rho_{j+\frac{1}{2}}^{n+1}, & u_{S,j}^n < 0, \end{cases} \qquad \rho_j^{[N],n+1} = \begin{cases} \rho_{j-\frac{1}{2}}^{n+1}, & u_{N,j}^n \geq 0, \\ \rho_{j+\frac{1}{2}}^{n+1}, & u_{N,j}^n < 0. \end{cases}$$

Equivalently, the limiting scheme can be written in matrix form as

$$M^n \boldsymbol{\rho}^{n+1} = \boldsymbol{\rho}^n.$$

The tridiagonal matrix M^n has entries

$$\begin{aligned}
m_{j,j-1}^n &= -D_{\rho,h}\frac{\Delta t}{(\Delta x)^2} - \frac{\Delta t}{\Delta x}(u_{S,j}^n)^+ - \frac{\Delta t}{\Delta x}(u_{N,j}^n)^+, \\
m_{j,j}^n &= 1 + 2D_{\rho,h}\frac{\Delta t}{(\Delta x)^2} + \frac{\Delta t}{\Delta x}\left[(u_{S,j+1}^n)^+ + (u_{S,j}^n)^-\right] + \frac{\Delta t}{\Delta x}\left[(u_{N,j+1}^n)^+ + (u_{N,j}^n)^-\right], \\
m_{j,j+1}^n &= -D_{\rho,h}\frac{\Delta t}{(\Delta x)^2} - \frac{\Delta t}{\Delta x}(u_{S,j+1}^n)^- - \frac{\Delta t}{\Delta x}(u_{N,j+1}^n)^-.
\end{aligned}$$

Here $(a)^+ = \max\{a, 0\}$ and $(a)^- = \max\{-a, 0\}$. The boundary rows of M^n are modified according to the homogeneous no-flux boundary conditions.

Proposition 5.1 (Mass conservation, positivity, and upper bound) *Assume that $D_{\rho,h} > 0$ and that the effective velocities satisfy*

$$|u_{S,j}^n| \leq U_S, \qquad |u_{N,j}^n| \leq U_N$$

for all j and n. Suppose further that

$$\frac{\Delta t}{\Delta x} \leq \frac{1}{4(U_S + U_N)}. \tag{5.7}$$

If the initial data satisfy $\rho_j^0 \geq 0$ for all j, then the limiting scheme satisfies

$$\rho_j^n \geq 0, \qquad \forall j,\ n \geq 0.$$

Moreover, under the homogeneous no-flux boundary conditions, the discrete mass is conserved,

$$\|\boldsymbol{\rho}^{n+1}\|_1 = \|\boldsymbol{\rho}^n\|_1,$$

and the solution satisfies the uniform bound

$$0 \le \rho_j^n \le \bar{\rho}, \qquad \bar{\rho} = 2\|\boldsymbol{\rho}^0\|_1.$$

Proof *We first observe that*

$$m_{j,j-1}^n + m_{j,j}^n + m_{j,j+1}^n = 1 + \frac{\Delta t}{\Delta x}\left(u_{S,j+1}^n - u_{S,j}^n\right) + \frac{\Delta t}{\Delta x}\left(u_{N,j+1}^n - u_{N,j}^n\right) =: F_j^n.$$

Using the bounds on u_S^n and u_N^n, we have

$$F_j^n \ge 1 - 2(U_S + U_N)\frac{\Delta t}{\Delta x}.$$

Under (5.7), this gives

$$F_j^n \ge \frac{1}{2}.$$

The off-diagonal entries of M^n are nonpositive. Moreover, the above lower bound implies the required diagonal dominance under the CFL condition. Hence M^n is a nonsingular M-matrix and $(M^n)^{-1}$ is nonnegative. Therefore, if $\boldsymbol{\rho}^n \ge 0$, then

$$\boldsymbol{\rho}^{n+1} = (M^n)^{-1}\boldsymbol{\rho}^n \ge 0.$$

Mass conservation follows from the conservative form of the scheme and the homogeneous no-flux boundary conditions. More precisely,

$$\|\boldsymbol{\rho}^{n+1}\|_1 = \mathbf{1}^T\boldsymbol{\rho}^{n+1} = \mathbf{1}^T M^n \boldsymbol{\rho}^{n+1} = \mathbf{1}^T\boldsymbol{\rho}^n = \|\boldsymbol{\rho}^n\|_1,$$

where we used the relation $\mathbf{1}^T M^n = \mathbf{1}^T$.

Finally, let $\bar{\rho} = 2\|\boldsymbol{\rho}^0\|_1$. By mass conservation and positivity,

$$\rho_j^n \le \|\boldsymbol{\rho}^n\|_1 = \|\boldsymbol{\rho}^0\|_1 = \frac{1}{2}\bar{\rho}.$$

Since $F_j^n \ge 1/2$, we obtain

$$(M^n(\bar{\rho}\mathbf{1}))_j = F_j^n\bar{\rho} \ge \frac{1}{2}\bar{\rho} \ge \rho_j^n.$$

Using the nonnegativity of $(M^n)^{-1}$, we conclude that

$$\boldsymbol{\rho}^{n+1} = (M^n)^{-1}\boldsymbol{\rho}^n \le (M^n)^{-1}M^n(\bar{\rho}\mathbf{1}) = \bar{\rho}\mathbf{1}.$$

This proves the desired upper bound.

6 Numerical experiments

In this section, we present numerical experiments to illustrate the travelling pulse behavior of the model and to verify the asymptotic preserving property of the proposed micro-macro scheme. We first validate the macroscopic solver by comparing the numerical travelling pulse with the analytical travelling profile. We then study the dependence of the travelling speed on the chemotactic sensitivities. Finally, we compare the kinetic and macroscopic solutions for different values of ε.

Numerical setting. Unless otherwise stated, we consider the one-dimensional computational domain $\Omega = (0, 150)$. For the profile and asymptotic preserving tests, we use

$$\Delta x = 0.1, \qquad \Delta t = 10^{-3}.$$

The velocity domain is $V = [-1, 1]$. It is discretized by a uniform trapezoidal quadrature with $\Delta v = 0.02$. We denote the discrete velocity nodes and weights by $\{v_k, \omega_k\}$, and use

$$\langle h \rangle_h = \sum_k \omega_k h(v_k), \qquad |V|_h = \sum_k \omega_k.$$

The discrete diffusion coefficient in the limiting macroscopic equation is

$$D_{\rho,h} = \frac{\langle v^2 \rangle_h}{|V|_h}.$$

For the default velocity grid with $\Delta v = 0.02$, this gives $D_{\rho,h} = 0.3334$.

Homogeneous no-flux boundary conditions are imposed for the density and the chemical variables. Since the computational domain is sufficiently large, the travelling pulse does not interact with the boundary during the time intervals considered below.

The initial data are chosen so that cells are concentrated near the left side of the domain, with no chemoattractant signal and a constant nutrient level. More precisely, we take

$$\rho(0, x) = \rho_0 e^{-\lambda x}, \qquad S(0, x) = 0, \qquad N(0, x) = N_0,$$

where

$$\rho_0 = 1, \qquad \lambda = 3, \qquad N_0 = 10^3.$$

For the kinetic computations, the initial data are chosen consistently with the micro-macro decomposition,

$$f^\varepsilon(0, x, v) = \frac{\rho(0, x)}{|V|_h} + \varepsilon g(0, x, v), \qquad g(0, x, v) = 0.$$

The chemical parameters are

$$D_S = 2, \qquad D_N = 0, \qquad \alpha = 0.05, \qquad \beta = \gamma = 1.$$

Unless otherwise stated, the chemotactic sensitivities in the profile and AP tests are

$$\chi_S = 0.5, \qquad \chi_N = 1.1.$$

We use the sign response function

$$\phi(y) = -\operatorname{sign}(y).$$

In the computations, the response is multiplied by the normalization factor

$$s_h = \frac{|V|_h}{\langle |v| \rangle_h}.$$

With this normalization, the effective drift amplitudes in the macroscopic limit are exactly χ_S and χ_N. For the macroscopic limiting runs, the response is evaluated with the coefficient of q_t set to zero. For the kinetic runs, the response is evaluated with the finite coefficient ε, namely in the form $\phi(\varepsilon q_t + v q_x)$. In the kinetic computations we also use the positivity-preserving post-processing described in Appendix B.

6.1 Macroscopic travelling pulse and analytical profile

We first validate the macroscopic solver by comparing the numerical travelling pulse with the analytical travelling profile. Let x_p be the peak position of the numerical density and set

$$z = x - x_p, \qquad z_- = \min\{z, 0\}, \qquad z_+ = \max\{z, 0\}.$$

The analytical travelling profile is written as

$$\rho_{\mathrm{tw}}(x) = \rho_p \exp\left(\lambda_L z_- + \lambda_R z_+\right), \tag{6.1}$$

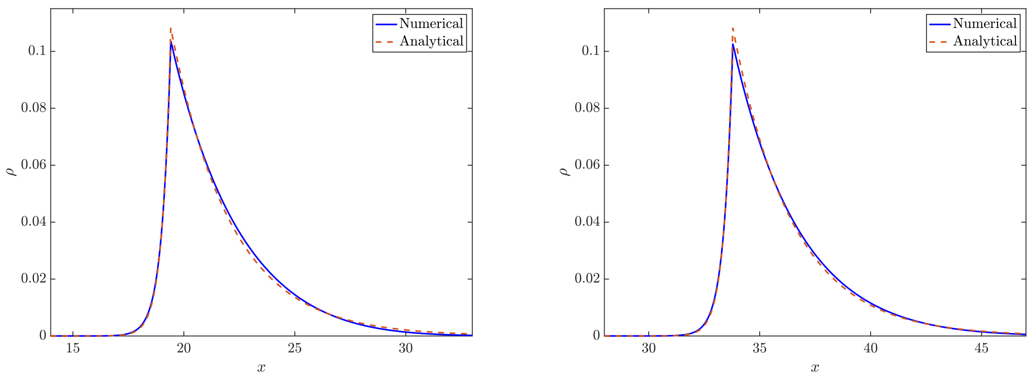
where

$$\lambda_L = \frac{\chi_S + \chi_N - \sigma}{D_{\rho,h}}, \qquad \lambda_R = \frac{\chi_N - \chi_S - \sigma}{D_{\rho,h}}.$$

Here $\lambda_L > 0$ and $\lambda_R < 0$, so that the profile decays on both sides of the peak. The amplitude ρ_p is chosen by discrete mass matching. More precisely, if $z_j = x_{j+\frac{1}{2}} - x_p$, then

$$\rho_p = \frac{\Delta x \sum_j \rho_j}{\Delta x \sum_j \exp\left(\lambda_L (z_j)_- + \lambda_R (z_j)_+\right)}.$$

Figure 1 compares the macroscopic numerical profiles with the analytical travelling profiles at two representative times. The peak position and amplitude of the analytical profile are chosen from the corresponding numerical solution. The close agreement confirms that the macroscopic solver captures the expected travelling wave structure. This macroscopic solution will be used as the reference solution in the kinetic-to-macroscopic comparison below.



(a) $t = 30$ (b) $t = 50$

Figure 1: Comparison between the macroscopic numerical profiles and the analytical travelling profiles at $t = 30$ and $t = 50$.

6.2 Dependence of the travelling speed on chemotactic sensitivities

We next investigate how the travelling speed depends on χ_S and χ_N. For each pair (χ_S, χ_N), the predicted travelling speed σ is determined by

$$\chi_N - \sigma = \chi_S \frac{\sigma}{\sqrt{4D_S\alpha + \sigma^2}}. \tag{6.2}$$

To estimate the travelling speed numerically, we track the center of mass of the density,

$$X_\rho(t_n) = \frac{\Delta x \sum_j x_{j+\frac{1}{2}} \rho^n_{j+\frac{1}{2}}}{\Delta x \sum_j \rho^n_{j+\frac{1}{2}}}.$$

The travelling speed is then estimated from the center-of-mass trajectory $X_\rho(t)$.

For the parameter scan in this subsection, we use a slightly coarser mesh to reduce the computational cost, i.e. $\Delta x = 0.2, \Delta v = 0.04, \Delta t = 10^{-2}$. The sensitivities χ_S, χ_N are uniformly sampled on $[0, 2] \times [0, 2]$. The wave speeds are evaluated at $t = 40$.

Figure 2 shows the dependence of the travelling speed on χ_S and χ_N. The left panel gives the speed predicted by (6.2), while the right panel shows the numerical speed obtained from the macroscopic model by tracking the center of mass. The two panels agree well in the parameter regime where a clear travelling pulse is formed. Some discrepancies are observed near the transition region where the nutrient bias is not sufficiently strong compared with the self-attractant bias. In this region, the pulse formation may require a longer transient time, and the finite-time center-of-mass estimate may underestimate the asymptotic travelling speed.

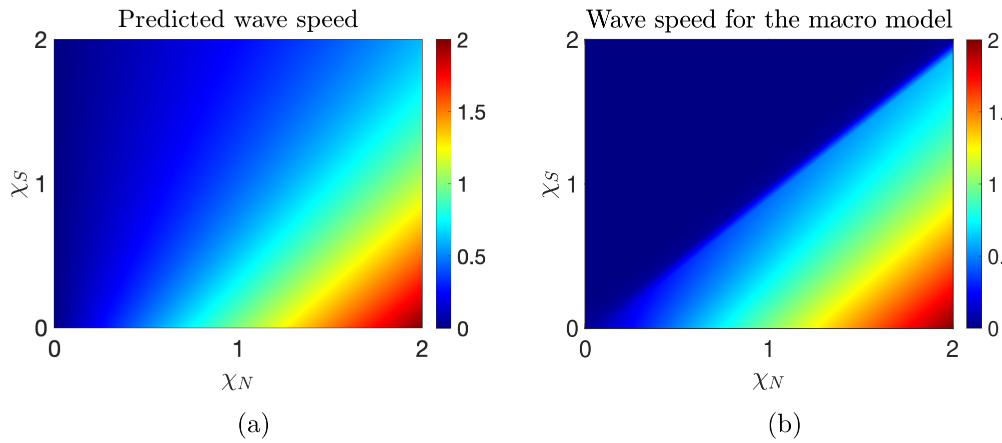


Figure 2: Dependence of the travelling speed on χ_S and χ_N. The left panel shows the speed predicted by (6.2), while the right panel shows the numerical speed obtained from the macroscopic model.

6.3 Asymptotic preserving convergence

We now compare the kinetic solution with the limiting macroscopic solution for different values of ε. The purpose of this test is to verify that, for fixed Δx, Δv, and Δt, the kinetic AP solution approaches the limiting macroscopic solution as ε decreases. We use the baseline parameter setting described above and consider

$$\varepsilon = 0.1, \qquad \varepsilon = 0.05, \qquad \varepsilon = 0.01.$$

Figure 3 compares the travelling density pulses obtained from the kinetic model and the limiting macroscopic model at several representative times. As ε decreases, the kinetic profiles become progressively closer to the macroscopic profile. In particular, for $\varepsilon = 0.01$, the kinetic and macroscopic solutions are already nearly indistinguishable at the plotted times. This provides direct evidence that the AP scheme captures the correct diffusive limit.

To quantify the convergence, we use the discrete L^2-error

$$\mathrm{err}_{\rho,\varepsilon}(t) = \left(\Delta x \sum_j \left| \rho^{\varepsilon}_{j+\frac{1}{2}}(t) - \rho_{j+\frac{1}{2}}(t) \right|^2 \right)^{1/2},$$

where ρ^{ε} is reconstructed from the kinetic solution and ρ denotes the solution of the limiting macroscopic model. Figure 4 plots the final-time error $\mathrm{err}_{\rho,\varepsilon}(50)$ for different values of ε. The error decreases monotonically as ε becomes smaller, confirming the convergence of the kinetic AP solution toward the macroscopic one.

Besides the density profile, it is also important to compare the travelling speed. The instantaneous speed is computed from the center-of-mass trajectory $X_\rho(t)$. Figure 5 shows

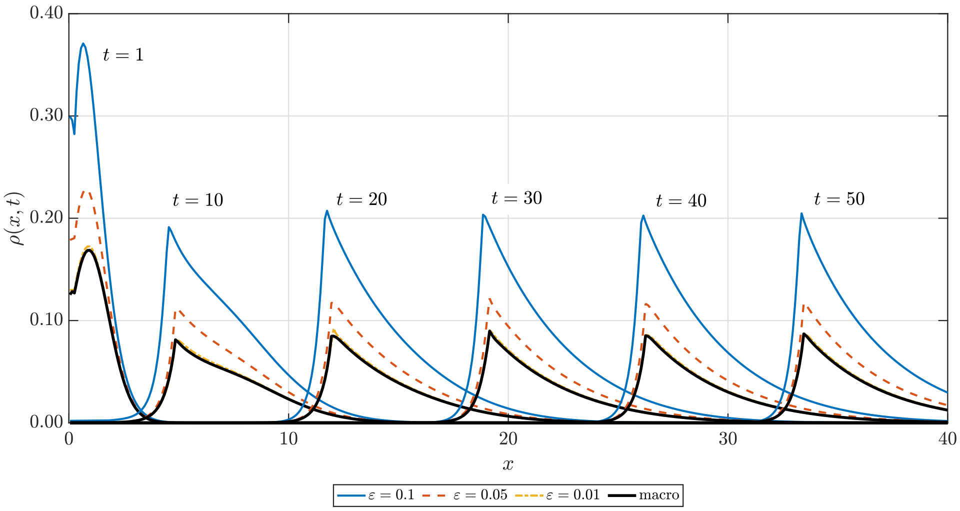


Figure 3: Comparison of travelling density pulses obtained from the kinetic model and the limiting macroscopic model at $t = 1, 10, 20, 30, 40, 50$. The propagation of the pulse is captured at both levels, and the kinetic profiles approach the macroscopic profiles as ε decreases.

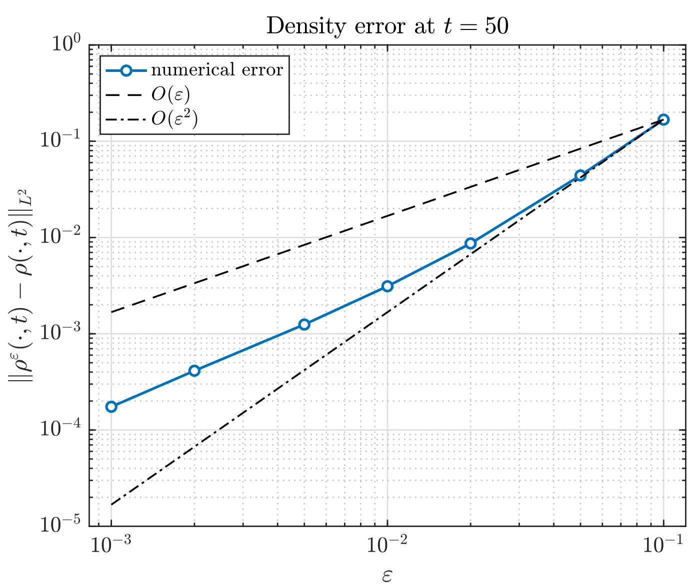


Figure 4: Final-time density error $\mathrm{err}_{\rho,\varepsilon}(50)$ versus ε. The reference lines $O(\varepsilon)$ and $O(\varepsilon^2)$ are included for comparison.

the speeds obtained from the macroscopic solution and from the kinetic solutions for different values of ε. The horizontal dashed line indicates the predicted value

$$\sigma_{\mathrm{pred}} = 0.724,$$

obtained from (6.2). After a short transient regime, all curves approach a common plateau. Moreover, the kinetic wave speed becomes closer to the macroscopic one as ε decreases.

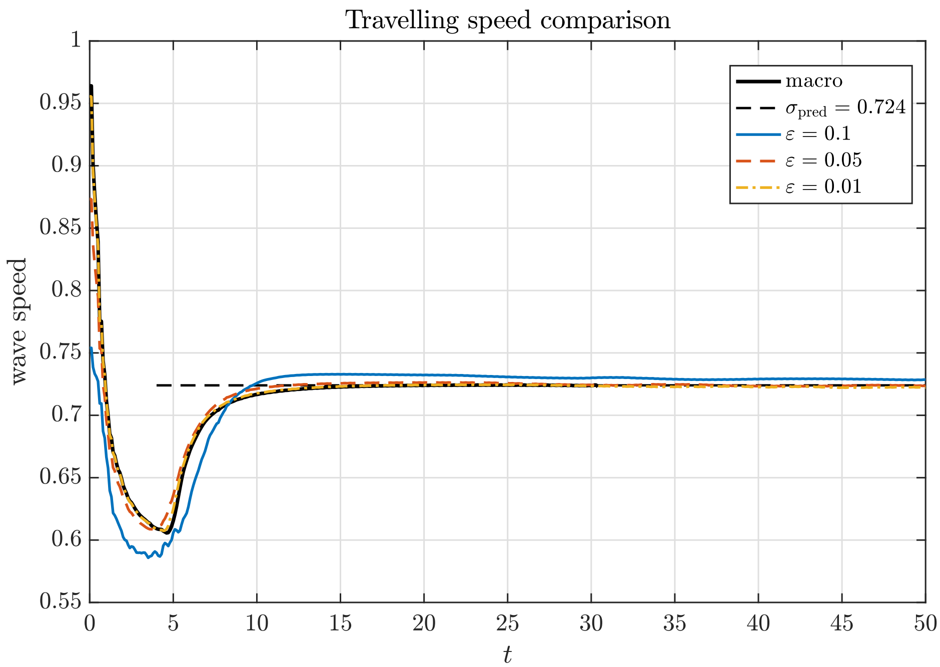


Figure 5: Travelling speed comparison between the macroscopic model and the kinetic model for different values of ε. The horizontal dashed line indicates the macroscopic predicted speed $\sigma_{\mathrm{macro}} \approx 0.724$, obtained from (6.2).

Overall, these results show that the proposed AP scheme captures the travelling pulse profiles at the kinetic level and converges to the limiting macroscopic dynamics, both in density profiles and in propagation speed, as ε decreases.

A Tumbling operator

To better understand these values, we will explain the intuition behind the signs in each expression of T.

- $T_-^+ = 1 - \chi_S - \chi_N$: In this particular case, the $+$ tells us that $v > c$, i.e. the bacteria of this group are moving faster than the wave, while the $-$ corresponds to $z < 0$ which means that bacteria are at the left side of the peak of S concentration. Therefore

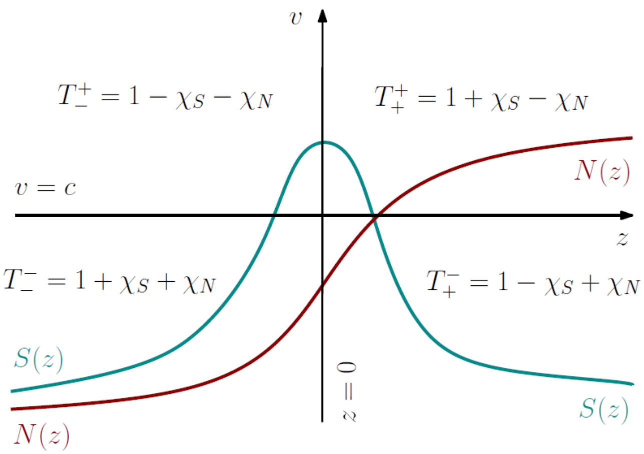


Figure 6: In this figure there are plotted the profiles of S and N on the plane defined by the axes $z = 0$ and $v = c$. In each quadrant we have the corresponding value of the tumbling rate.

we are in the situation where bacteria move towards the maximum concentration of S and also towards higher concentration of N, since it is a favourable motion we do not want to change it that much so we need the tumbling rate as small as possible, so T_-^+ is minimum.

- $T_+^+ = 1 + \chi_S - \chi_N$: Here, we also have $v > c$ which indicates that the bacteria are moving faster than the wave (to the right). As far as the nutrient is concerned, we are going in the right direction, thus, χ_N does not contribute to more tumbling, on the contrary, wants less tumbling, and that is why we have $-\chi_N$ as well as in the case above. On the other hand, the subscript $+$ shows that $z > 0$, i.e. we are in the right side of the peak of S. So, as far as the S is concerned, bacteria move away from the higher concentration of S (not favourable for S). Consequently, the factor χ_S does contribute to a higher tumbling rate to revert this situation, that is why we have a positive $+\chi_S$.

- $T_-^- = 1 + \chi_S + \chi_N$: In the third setup we have $v < c$ which means that we are moving slower or even in the opposite direction to the wave, and $z < 0$ indicating bacteria are in the left side of the peak of S. We are in the worst scenario because bacteria move away (or goes slowly) towards the peak of S and from the higher concentration of N too. Taking this into account the best is to tumble more to see if we can rectify this situation. Hence, the T is maximal: both factors contribute to have more tumbling.

- $T_+^- = 1 - \chi_S + \chi_N$: Finally, we have the setup where the $-\chi_S$ does not contribute to more tumbling because, since bacteria are at the right side of the peak, moving backwards or slower it is fine for S. Oppositely, it is not a good situation for N, so the factor $+\chi_N$ does contribute to more tumbling.

B Post-processing for positivity preservation

In this appendix, we describe the positivity-preserving post-processing used after each time step. Let $\tilde{\rho}^{n+1}_{j+\frac{1}{2}}$ and $\tilde{g}^{n+1}_{j,k}$ be the numerical solution obtained from (5.1). We use the discrete velocity average

$$\langle h \rangle_h = \sum_{k=1}^{N_v} \omega_k h(v_k), \qquad \sum_{k=1}^{N_v} \omega_k = 1, \qquad \langle \psi_0 \rangle_h = 1.$$

We also assume that the microscopic variable satisfies $\langle \tilde{g}^{n+1}_{j,\cdot} \rangle_h = 0$.

We first reconstruct the cell-centered microscopic component by

$$\tilde{\tilde{g}}^{n+1}_{j+\frac{1}{2},k} = \frac{\tilde{g}^{n+1}_{j,k} + \tilde{g}^{n+1}_{j+1,k}}{2}.$$

Then the corresponding distribution is given by

$$\tilde{f}^{n+1}_{j+\frac{1}{2},k} = \tilde{\rho}^{n+1}_{j+\frac{1}{2}} \psi_0(v_k) + \epsilon \tilde{\tilde{g}}^{n+1}_{j+\frac{1}{2},k}.$$

By the normalization of ψ_0 and the zero-mean property of g, we have

$$\langle \tilde{f}^{n+1}_{j+\frac{1}{2},\cdot} \rangle_h = \tilde{\rho}^{n+1}_{j+\frac{1}{2}}.$$

For each fixed cell $j + \frac{1}{2}$, define

$$Q^{n+1}_{j+\frac{1}{2},k} = \tilde{\rho}^{n+1}_{j+\frac{1}{2}} \psi_0(v_k).$$

If $\tilde{\rho}^{n+1}_{j+\frac{1}{2}} > 0$, we set

$$\theta_{j+\frac{1}{2}} = \min \left\{ 1, \min_{\tilde{f}^{n+1}_{j+\frac{1}{2},k} < 0} \frac{Q^{n+1}_{j+\frac{1}{2},k}}{Q^{n+1}_{j+\frac{1}{2},k} - \tilde{f}^{n+1}_{j+\frac{1}{2},k}} \right\},$$

where the inner minimum is omitted if $\tilde{f}^{n+1}_{j+\frac{1}{2},k} \geq 0$ for all k. We then define

$$f^{n+1,*}_{j+\frac{1}{2},k} = Q^{n+1}_{j+\frac{1}{2},k} + \theta_{j+\frac{1}{2}} \left(\tilde{f}^{n+1}_{j+\frac{1}{2},k} - Q^{n+1}_{j+\frac{1}{2},k} \right),$$

and set

$$\rho^{n+1,*}_{j+\frac{1}{2}} = \tilde{\rho}^{n+1}_{j+\frac{1}{2}}, \qquad \bar{g}^{n+1,*}_{j+\frac{1}{2},k} = \frac{f^{n+1,*}_{j+\frac{1}{2},k} - \rho^{n+1,*}_{j+\frac{1}{2}} \psi_0(v_k)}{\epsilon}.$$

This scaling keeps the cell average unchanged and gives

$$f_{j+\frac{1}{2},k}^{n+1,*} \geq 0, \qquad \langle \bar{g}_{j+\frac{1}{2},\cdot}^{n+1,*} \rangle_h = 0.$$

If $\tilde{\rho}_{j+\frac{1}{2}}^{n+1} \leq 0$, positivity cannot be recovered without changing the cell average. In this case, we set

$$\rho_{j+\frac{1}{2}}^{n+1,*} = 0, \qquad f_{j+\frac{1}{2},k}^{n+1,*} = 0, \qquad \bar{g}_{j+\frac{1}{2},k}^{n+1,*} = 0.$$

The above cellwise correction may change the total mass. We therefore apply a global rescaling. Let

$$M_h^{n+1} = \Delta x \sum_j \tilde{\rho}_{j+\frac{1}{2}}^{n+1}, \qquad M_h^{n+1,*} = \Delta x \sum_j \rho_{j+\frac{1}{2}}^{n+1,*}.$$

When $M_h^{n+1,*} > 0$, define

$$\alpha = \frac{M_h^{n+1}}{M_h^{n+1,*}}.$$

The final limited variables are given by

$$\rho_{j+\frac{1}{2}}^{n+1} = \alpha \rho_{j+\frac{1}{2}}^{n+1,*}, \qquad \bar{g}_{j+\frac{1}{2},k}^{n+1} = \alpha \bar{g}_{j+\frac{1}{2},k}^{n+1,*}.$$

Equivalently,

$$f_{j+\frac{1}{2},k}^{n+1} = \rho_{j+\frac{1}{2}}^{n+1} \psi_0(v_k) + \epsilon \bar{g}_{j+\frac{1}{2},k}^{n+1} = \alpha f_{j+\frac{1}{2},k}^{n+1,*}.$$

Therefore the post-processed distribution satisfies

$$f_{j+\frac{1}{2},k}^{n+1} \geq 0 \quad \text{for all } j, k,$$

and the total mass is conserved in the sense that

$$\Delta x \sum_j \rho_{j+\frac{1}{2}}^{n+1} = \Delta x \sum_j \tilde{\rho}_{j+\frac{1}{2}}^{n+1}.$$

Moreover,

$$\langle \bar{g}_{j+\frac{1}{2},\cdot}^{n+1} \rangle_h = 0.$$